

National Health and Nutrition Examination Survey

August 2021-August 2023 Data Documentation, Codebook, and Frequencies

Volatile Toxicant (VTQ_L)

Data File: VTQ_L.xpt

First Published: December 2025

Last Revised: NA

Component Description

The volatile toxicant questionnaire (variable name prefix VTQ) includes data about the participants home, activities, amount of time spent in various locations and exposure to different chemicals over the past 48 hours. This questionnaire data can be used in conjunction with the volatile organic compound (VOC) laboratory dataset in the NHANES August 2021 – August 2023 Laboratory Data section.

VOCs are a large group of chemicals that have been used as solvents, degreasers, and cleaning agents in industry and consumer products. Many VOCs were found to contaminate ground water and drinking water sources, and because of human health concerns, have been banned or restricted from most uses.

Eligible Sample

Participants aged 12 years and older from a one-half sample were eligible.

Interview Setting and Mode of Administration

The volatile toxicant questionnaire was asked in the mobile examination center (MEC), by trained phlebotomists, using the computer-assisted personal interview (CAPI) system.

Quality Assurance & Quality Control

VTQ data were systematically reviewed for consistency, data processing, and data input errors on a periodic basis during the data collection.

The CAPI system is programmed with built-in consistency checks to reduce data entry errors. CAPI also uses online help screens to assist interviewers in defining key terms used

Data Processing and Editing

Data were edited for completeness and consistency.

Analytic Notes

Please refer to the [NHANES Analytic Guidelines](#) and the on-line [NHANES Tutorial](#) for further details on the use of sample weights and other analytic issues.

Codebook and Frequencies

SEQN - Respondent sequence number

Variable Name:	SEQN
SAS Label:	Respondent sequence number
English Text:	Respondent sequence number.
Target:	Both males and females 12 YEARS - 150 YEARS

WTSVOC2Y - VOC Subsample Weight

Variable Name: WTSVOC2Y

SAS Label: VOC Subsample Weight

English Text: VOC Subsample Weight

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
11386.962716 to 543590.56928	Range of Values	2960	2960	
0	No Lab Result	658	3618	End of Section
.	Missing	0	3618	

VTQ231A - Use moth balls or toilet deodorizers?

Variable Name: VTQ231A

SAS Label: Use moth balls or toilet deodorizers?

English Text: {Do you/Does she/Does he} currently use moth balls, moth crystals or toilet bowl deodorizers inside {your/her/his} home?

English Instructions: HELP SCREEN SHOULD READ: Some toilet bowl deodorizers clip onto the toilet rim, other, such as deodorant blocks and gels, are placed inside the tank or hang inside the wall of the tank. Brand names include Bully, 2000 Flushes, Vanish, X-14, Ty-D-Bol, Toilet Duck, Clorox, Lime-A-Way, and Sno Bol.

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	252	252	
2	No	2665	2917	
7	Refused	0	2917	
9	Don't know	40	2957	
.	Missing	661	3618	

VTQ233A - Past 3 days inhale smoke for 10 minutes

Variable Name: VTQ233A

SAS Label: Past 3 days inhale smoke for 10 minutes

English Text: In the last three days, did {you/she/he} inhale smoke from any source for 10 or more minutes?

Target: Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	473	473	
2	No	2482	2955	
7	Refused	0	2955	
9	Don't know	2	2957	
.	Missing	661	3618	

VTQ233B - When did you last inhale smoke for 10min

Variable Name: VTQ233B**SAS Label:** When did you last inhale smoke for 10min**English Text:** When did {you/she/he} last spend 10 or more minutes inhaling smoke?**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Today	252	252	
2	Yesterday	157	409	
3	More than 2 days	63	472	
7	Refused	0	472	
9	Don't know	1	473	
.	Missing	3145	3618	

VTQ244A - Did you pump gas into your car?**Variable Name:** VTQ244A**SAS Label:** Did you pump gas into your car?**English Text:** In the last 48 hours, did {you/she/he} pump gas into a car or other motor vehicle {yourself/herself/ himself}?**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	638	638	
2	No	2311	2949	VTQ261A
7	Refused	0	2949	VTQ261A
9	Don't know	8	2957	VTQ261A
.	Missing	661	3618	

VTD244B - Long ago did you pump gas in your car?**Variable Name:** VTD244B**SAS Label:** Long ago did you pump gas in your car?**English Text:** How long ago, in hours, did {you/she/he} pump gas into a car or other motor vehicle {yourself/herself/ himself}?**English Instructions:** HOURS HARD EDIT: Range - 1 - 48**Target:** Both males and females 12 YEARS - 150 YEARS**Hard Edits:** 1 to 48

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 48	Range of Values	628	628	
0	Less than one hour	10	638	
77	Refused	0	638	
99	Don't know	0	638	
.	Missing	2980	3618	

VTQ261A - In last 48 hrs use dry cleaning solvent?**Variable Name:** VTQ261A**SAS Label:** In last 48 hrs use dry cleaning solvent?**English Text:** In the last 48 hours, did {you/she/he} use dry cleaning solvents, visit a dry cleaning shop or wear clothes that had been dry-cleaned within the last week?**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	50	50	
2	No	2899	2949	VTQ271A
7	Refused	0	2949	VTQ271A
9	Don't know	8	2957	VTQ271A
.	Missing	661	3618	

VTQ261B - Long since use dry cleaning solvents?

Variable Name: VTD261B

SAS Label: Long since use dry cleaning solvents?

English Text: How long ago, in hours, has it been since {you/she/he} used dry cleaning solvents, visited a dry cleaning shop or wore clothes that had been dry-cleaned within the last week?

English Instructions: HARD EDIT: Range - 1 - 48 HOURS

Target: Both males and females 12 YEARS - 150 YEARS

Hard Edits: 1 to 48

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 48	Range of Values	49	49	
0	Less than one hour	0	49	
77	Refused	0	49	
99	Don't know	0	49	
.	Missing	3569	3618	

VTQ271A - In last 48 hrs, bathe for 5+ minutes?**Variable Name:** VTQ271A**SAS Label:** In last 48 hrs, bathe for 5+ minutes?**English Text:** In the last 48 hours, did {you/she/he} take a hot shower or bath for five minutes or longer?**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	2286	2286	
2	No	666	2952	VTQ281A
7	Refused	0	2952	VTQ281A
9	Don't know	5	2957	VTQ281A
.	Missing	661	3618	

VTD271B - How long, in hrs, since you bathed?

Variable Name: VTD271B**SAS Label:** How long, in hrs, since you bathed?**English Text:** How long ago, in hours, has it been since {your/SP's} last shower or hot bath?**English Instructions:** Hours**Target:** Both males and females 12 YEARS - 150 YEARS**Hard Edits:** 1 to 48

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 48	Range of Values	2285	2285	
0	Less than one hour	0	2285	
77	Refused	0	2285	
99	Don't know	0	2285	
.	Missing	1333	3618	

VTQ281A - In past 48 hrs breathe paint fumes?**Variable Name:** VTQ281A**SAS Label:** In past 48 hrs breathe paint fumes?**English Text:** In the last 48 hours, did {you/she/he} breathe fumes from freshly painted indoor surfaces, paints, paint thinner, or varnish?**Target:** Both males and females 12 YEARS - 150 YEARS

Code or Value	Value Description	Count	Cumulative	Skip to Item
1	Yes	80	80	
2	No	2872	2952	End of Section
7	Refused	0	2952	End of Section
9	Don't know	4	2956	End of Section
.	Missing	662	3618	

VTD281B - How long, since breathed paint fumes?

Variable Name: VTD281B

SAS Label: How long, since breathed paint fumes?

English Text: How long ago, in hours, has it been since {you/she/he} breathed fumes from freshly painted indoor surfaces, paints, paint thinner, or varnish?

English Instructions: HARD EDIT: Range - 1 - 48 HOURS

Target: Both males and females 12 YEARS - 150 YEARS

Hard Edits: 1 to 48

Code or Value	Value Description	Count	Cumulative	Skip to Item
1 to 48	Range of Values	80	80	
0	Less than one hour	0	80	
77	Refused	0	80	
99	Don't know	0	80	
.	Missing	3538	3618	